Job Shop Scheduling Tutorial

You should do this tutorial using Google Colaboratory and using Google OR-Tools.

Note you will need to run the following first to install the ortools python package.

```
!pip install ortools
```

Exercises

Solve the following problems by adapting either the Google OR-Tools code or the Pyomo MILP code from the notes. Q1–Q3 can also be done in the browser, with nothing to install: adapt the Pyomo code in the *Job Shop Scheduling Notes* notebook (Week 7, `1 Lecture/Notes`).

Sketch out your solution and check that it is indeed feasible and that it appears to be optimal. Remember you cannot check optimality (if done correctly it will be optimal), but you can at least eyeball the solution for reasonableness.

1. Job Shop Scheduling Problem:

Jobs	Operations	Sequence on Machines	Process Time on Machines
J_0	O_{01}, O_{02}, O_{03}	M_0, M_1, M_2	3,2,2
J_1	O_{11}, O_{12}, O_{13}	M_0, M_2, M_1	4,3,1
J_2	O_{21}, O_{22}, O_{23}	M_1, M_0, M_2	2,3,2

2. Job Shop Scheduling Problem:

Jobs	Operations	Sequence on Machines	Process Time on Machines
J_1	$O_{11}, O_{12}, O_{13}, O_{14}$	A, B, C, D	3,2,4,3
J_2	O_{21}, O_{22}, O_{23}	B, A, D	5,3,2
J_3	$O_{31}, O_{32}, O_{33}, O_{34}$	C, D, A, B	2,1,4,4
J_4	O_{41}, O_{42}, O_{43}	D, C, B	4,3,5

3. Job Shop Scheduling Problem:

Jobs	Operations	Sequence on Machines	Process Time on Machines
J_A	$O_{A1}, O_{A2}, O_{A3}, O_{A4}, O_{A5}$	M_1, M_2, M_3, M_4, M_5	4,2,3,5,2
J_B	$O_{B1}, O_{B2}, O_{B3}, O_{B4}$	M_3, M_4, M_2, M_5	2,1,3,4
J_C	$O_{C1}, O_{C2}, O_{C3}, O_{C4}$	M_1, M_2, M_3, M_4	3,4,2,3

4. N-queens Problem:

Read through the **N-queens problem** on the Google OR-Tools site.

Implement the solution in Google Colaboratory and make sure you understand what the model is doing and how it translates to the Python code.

5. Employee Scheduling Problem:

Read through the **Employee Scheduling Problem** on the Google OR-Tools site.

Implement the solution in Google Colaboratory and make sure you understand what the model is doing and how it translates to the Python code.